

Shreyas Peyyeti

AI Engineer

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Aspiring AI Engineer with a strong focus on Generative AI and intelligent automation. Passionate about building production-ready agentic systems that solve real-world business problems. Keen on designing scalable GenAI architectures and exploring cutting-edge techniques to push the boundaries of AI applications.

SKILLS

AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, OpenCV, spaCy, NLTK, YOLO

Gen AI Frameworks/protocols: LangChain, LangGraph, Hugging Face Transformers, Google ADK, Crew AI, MCP, Google A2A

Development: FastAPI, Flask, REST APIs, Git, Docker

Databases: PostgreSQL, Microsoft SQL Server, Qdrant, FAISS, Chroma

PROFESSIONAL EXPERIENCE

Aragen Life Sciences

Agentic AI Intern | LangChain, QdrantDB, Microsoft SQL Server, REST APIs, Ollama

April 2025 – Present

- Developing **MAP**, a production GenAI platform for transcript summarization, RAG Q&A, action extraction, and sentiment analysis serving 2000+ users, leveraging **Qdrant VectorDB** and **Microsoft SQL Server** for robust data storage solutions.
- Built production-ready agentic chatbot using **Llama3.2** and **LangGraph** for automated conversation processing.
- Developing **MCP**-based AI agents for autonomous meeting workflow automation and intelligent data processing pipelines.
- Driving innovation by architecting and implementing agentic use cases for Aragen's strategic projects, enhancing operational efficiency and user engagement through advanced AI technologies.

DeepStack Solutions

ML Intern | YOLOv8, OpenCV, Python, Stockfish

June 2024 – Aug 2024

- Developed real-time chessboard position detection system using **YOLOv8-OB** object detection, achieving **99.5% mAP@50** and **98.3% mAP@50-95** for precise piece localization.
- Engineered advanced image preprocessing workflows with **OpenCV**, including perspective correction and board enhancement algorithms, improving inference speed by 4.6x
- Integrated Stockfish chess engine API to suggest optimal next moves based on detected board states, enabling automated gameplay analysis and strategy recommendations

PROJECTS

Clinical Reasoning LLM Fine-Tuning | LoRA, Unsloth, PyTorch

Jan 2025- Feb 2025

- Fine-tuned the 8B **DeepSeek-R1** model using **LoRA (via Unsloth)** on the **Medical-CoT** dataset for clinical reasoning tasks.
- Achieved **1.45 training loss** with mixed-precision optimization on **T4 GPU**.
- Implemented **TRL's SFTTrainer** with structured prompt engineering and integrated **WandB** for comprehensive training monitoring.

PennyPal | LangGraph, Ollama, FastAPI, PostgreSQL, OpenCV

June 2025- Present

- Developing intelligent expense splitting application with receipt-based data extraction using **Llama3.2-vision** and **OpenCV** for real-time OCR processing and text recognition
- Building conversational AI agent using **LangGraph** multi-agent framework for natural language expense management and automated database interactions
- Designed robust **PostgreSQL** backend with **FastAPI** REST architecture, JWT authentication, and modular microservices for expense creation, tracking, and settlement processing

RESEARCH WORK

Alzheimer's Detection using Deep Learning | CNN, ResNet, CBAM, ADNI Dataset

Jan 2024 – May 2024

- Built an AI-powered plant identification system using **MobileNetV2** with **92% accuracy**.
- Implemented **OpenCV** preprocessing pipeline for real-time mobile image recognition and classification
- Integrated **Flask** API with **Gemini LLM** for generating plant-based medicinal insights and recommendations

Medical Text Simplification System | SimpleT5, MeDAL

Aug 2024 – Dec 2024

- Fine-tuned **T5-base transformer** using **SimpleT5** on the **MeDAL dataset** to simplify medical abbreviations and acronyms.
- Developed a translation pipeline with custom preprocessing algorithms and position-aware word replacement.
- Achieved **BLEU score of 0.87** and **ROUGE-L score of 0.84** on medical text translation tasks.

EDUCATION

Bachelor of Technology (CS Engg), Mahindra École Centrale, CGPA: 7.6/10

Aug 2021 – July 2025

Intermediate, Page Jr. College, Hyderabad, Percentage: 96.6%

June 2019 – May 2021

10th Grade, Future Kid's School (ICSE), Hyderabad, Percentage: 84.5%

May 2011 – June 2019